Studying Ground-State Energy Estimation Through Quantum Sampling

Qiskit Fall Fest 2026

Ashiru Muhd Idris Hamzat Farouk Adegoke

Team Mensia

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Project Plan

Investigate how computational-basis measurements from a quantum circuit can be used to construct a sampled subspace for estimating the ground-state energy of a small quantum spin system.

Background and Mathematical Model

The ground-state energy is the lowest eigenvalue of a Hamiltonian operator:

$$H|\psi_0\rangle = E_0|\psi_0\rangle.$$

We study the transverse-field Ising chain:

$$H = -\sum_{i=0}^{N-2} Z_i Z_{i+1} - g \sum_{i=0}^{N-1} X_i.$$

- $N = 8$ qubits.
- Transverse-field strength: $g = 0.4$.
- Open boundary conditions.
- X_i and Z_i are Pauli operators acting on qubit i.

Objective

Compare the sampled-subspace energy with an exact classical reference and a random-subspace baseline.

Methodology and Qiskit Workflow

1. **Construct the Hamiltonian:** Represent the Ising model using Qiskit's `SparsePauliOp`.
2. **Calculate the reference:** Use classical numerical diagonalization to obtain the lowest eigenvalue and corresponding eigenvector.
3. **Prepare and measure:** Prepare a circuit using the reference statevector and collect computational-basis samples using Qiskit.
4. **Build the sampled subspace:** Identify distinct bitstrings obtained from measurement.
5. **Compare energies:** Evaluate the sampled-subspace estimates at increasing shot counts and compare them with random-subspace results.

Important experimental limitation

The reference ground-state vector is calculated classically before circuit preparation. This is a controlled sampling demonstration, not an independent quantum solution to the ground-state problem.

Experimental Results

Exact classical reference:

$$E_0 \approx -7.4055913441.$$

Shots	Unique states	Sampled energy	Random energy
16	2	-7.000000	-2.213333
32	4	-7.077033	-2.630555
64	9	-7.152313	-3.632943
128	11	-7.219976	-4.362250
256	16	-7.258384	-4.497980
512	25	-7.356189	-4.897585
1024	31	-7.372454	-5.322903
2048	42	-7.384186	-5.727132
4096	50	-7.386607	-5.957571

Observation: At 4096 shots, the sampled-subspace energy is approximately -7.386607, about 0.018984 above the exact reference energy. The reported sampled energy is lower than the random-subspace energy at every listed shot count.

Discussion, Limitations, and Conclusion

Main findings

- Increasing the shot count improves the reported sampled-subspace energy estimate toward the exact reference.
- The number of distinct observed bitstrings increased from 2 at 16 shots to 50 at 4096 shots in the recorded run.
- The random-subspace baseline produced less accurate energies in the reported experiment.

Limitations

- The statevector was obtained by classical diagonalization.
- The demonstration does not establish quantum advantage.
- Results depend on the sampling procedure and the definition of the sampled subspace.

Conclusion

The experiment illustrates how measurement samples can support subspace-based energy estimation. Further work should prepare states without using the exact ground-state vector and test the method on quantum hardware or a more realistic noisy simulation.

Declaration of Assistance and Resources

Project resources

- Qiskit for quantum-circuit construction and sampling.
- Python and NumPy for numerical computation and classical reference calculations.
- The Qiskit Fall Fest 2026 challenge materials and relevant Qiskit documentation.

AI assistance

ChatGPT was used to support project planning, explain mathematical and quantum-computing concepts, assist with code organization and debugging, and help prepare the presentation and written descriptions.

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